

6(a)	- consider <u>all</u> mps to award if response clearly addresses events in the left side - mps 1,2,6,7,8 must be in the context of the left side of the heart - otherwise allow mp 3, 4, and 5 to max 3 – these mark points are inset to the right any five from: 1 blood enters (heart) via / in, pulmonary vein(s) ; 2 blood exits heart to aorta ; R aorta 3 blood flows from atrium to ventricle ; 4 atrial, systole / contraction, then, ventricular, systole / contraction ; A atrium contracts then ventricle contracts 5 pressure in ventricle, increases / gets higher ; I high pressure 6 bicuspid / (left) atrioventricular valve closes ; A mitral valve R if in context of events in the atrium 7 semilunar / aortic, valve opens ; ecf from aorta in mp 2 R if in context of events in the atrium 8 bicuspid / (left) atrioventricular, valve closes when (blood) pressure in ventricle higher than atrium or semilunar / aortic, valve opens when (blood) pressure in ventricle higher than aorta ; A aortic valve	5
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} unclear
attempts / mixed
right and left
sides

Question	Answer	Marks
6(b)(i)	<i>if no marks gained allow one mark for four, polypeptides / polypeptide chains</i> two, (identical) alpha / α, chains / globins, <u>and</u> two, (identical) beta / β, chains / globins ; A subunits / polypeptides, for chains R beta-pleated sheets / globulin <i>plus one from;</i> each, chain / globin / polypeptide, incorporates / contains / AW, a <u>haem</u> (group) ; A four haem groups A molecule for polypeptide if mp1 gained A ferrous ion / Fe^{2+} / iron (ion / atom), within porphyrin (ring) for haem group A prosthetic group with, ferrous ion / Fe^{2+} / iron (ion / atom) hydrogen bonds / hydrophobic interactions / ionic bonds, between, chains / polypeptides / globins ; <i>context is interactions between chains</i> R if disulfide bond also stated	2
6(b)(ii)	oxygen binds to, iron ion / ferrous ion / Fe^{2+} ; A oxygen bonds / oxygen forms a bond, <i>only if</i>, iron ion / ferrous ion / Fe^{2+} stated A oxygen binds to, iron / iron atom / haem (group) R if more than one oxygen molecule binds to one iron ion detail of saturated ; e.g. total of 4 oxygen molecules / 8 oxygen atoms (bind per haemoglobin molecule) each, chain / globin / polypeptide, binds one oxygen molecule R bonds <i>ref. to</i> cooperative binding / allostery A description <i>ref. to</i> (fully saturated molecule is) in, high partial pressures of oxygen / high concentration of oxygen ; <i>ref. to</i> T / tense, state to, R / relaxed, state	2